GEOMETRIC IRREDUCIBILITY IS EQUIVALENT TO IRREDUCIBILITY OF THE SELF PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the function field criterion; no step is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-irreducible-self-product is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Behaviour of irreducible components under the self product and base change. Same chapter, Lemma lemma-bijection-irreducible-components, invoked at proof lines 1493-1495. Supplies the easy direction from geometric irreducibility to irreducibility of the self product.
- Criterion for geometric irreducibility through the function field. Same chapter, Lemma lemma-geometrically-irreducible-function-field, invoked at proof lines 1519-1520. Supplies that having no separably algebraic elements of the function field outside k implies geometric irreducibility.
- A finite separable extension E of k has E tensor_k E a product of fields of dimension dim_k(E). Standard field theory, used at proof lines 1513-1517. Supplies the contradictory decomposition of E tensor_k E into at least two fields.

Source reference. Original lemma 038F: sources/stacks-varieties/pinned-original-source.tex, source lines 1380–1380. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 054Q: sources/stacks-varieties/pinned-original-source.tex, source lines 1446–1446. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *A scheme X over a field k is geometrically irreducible over k if and only if $X \times_k X$ is irreducible.*

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. If X is geometrically irreducible over k, then $X \times_k X$ is irreducible by Lemma 038F.

Conversely, suppose that $X \times_k X$ is irreducible. Replacing X by the underlying reduced scheme, we can assume that X is reduced without changing the problem. Since the first projection $X \times_k X \to X$ is surjective, X is also irreducible, hence integral. Suppose that there is an element α in the function field $k(X)$ that is separably algebraic over k but not in k. Let $E = k(\alpha)$.

Then there is a nonempty affine open subscheme U of X such that $k \subset E \subset O(U)$, and so $E \otimes_k E \subset O(U) \otimes_k O(U) = O(U \times_k U)$. Thus we have a dominant morphism $U \times_k U \to \mathrm{Spec}(E \otimes_k E)$. Since $X \times_k X$ is irreducible, so is $U \times_k U$, and hence $\mathrm{Spec}(E \otimes_k E)$ is connected. If E is strictly bigger than k, then $E \otimes_k E$ is a product of fields including E as one factor. It has dimension $(\dim_k(E))^2 > \dim_k(E)$ as a k-vector space, and so $E \otimes_k E$ is a product of at least two fields, contradicting that $\mathrm{Spec}(E \otimes_k E)$ is connected. So in fact E is equal to k. By Lemma 054Q, X is geometrically irreducible over k. □